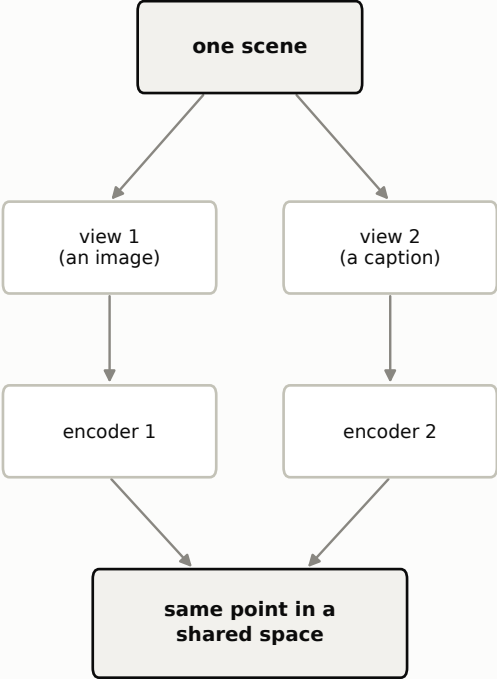
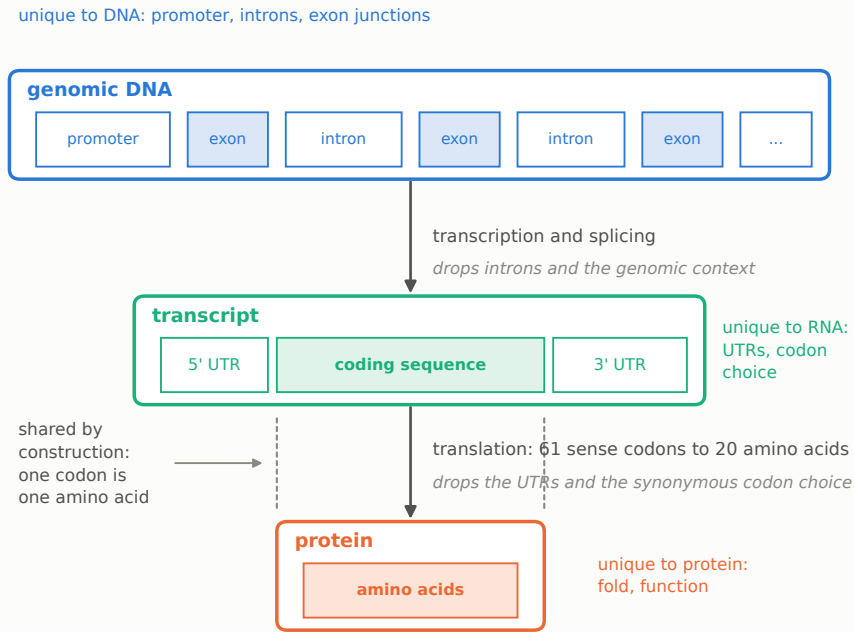


a Alignment in machine learning



Two views of one thing share almost everything, so alignment is a training target.

b The central dogma is not two views of one thing



The map runs one way and loses information at each step, so each molecule holds information the next one cannot have. Only the coding sequence and its translation are shared.

c What follows for alignment

Given by the genetic code

Positional alignment exists only on coding sequence, where every encoder's tokens sit on one codon grid. Stage 1: the grid resamples every stream; on GTEx no grid exists.

Measured on the shared part, and partial

Representational alignment is agreement that survives the composition control. Stage 1: CaLM still finds its protein partner 58% to 69% of the time; most DNA agreement is letter counts.

Not forced on the unique parts

An alignment loss would pull encoders toward what they share and erase what each one adds. Stage 1: translation erases GC3 (protein loses 0.26 to 0.30); on GTEx, DNA adds signal with Recall@1 at most 0.016.

Alignment in biology: agreement on what the central dogma shares, beyond composition, with nothing imposed on what it does not.